

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

**Title: Introduction to assembly program
structure and various arithmetic operations.**

Assembly Programming Lab
CSE 3106

Jashore University of Science and Technology

1 Objective(s)

- To provide an introduction to syntax and structure of assembly language.
- To implement various basic arithmetical operations.

2 Problem analysis

An assembly program basically divided into three sections. They are- (i) The Data section, (ii) The bss section and (iii) The text section. The Data section is used for initializing data and constants. This data doesn't change during runtime. The bss section is used for declaring variables and at last the text section contains actual code. In this section the actual program execution begins. Syntax of assembly language follows as- [Labels] mnemonic [Operands] [;Comments] in this format. For arithmetic operations ADD, SUB, MUL, DIV etc commands are used in assembly. In the given example, two numbers

2.1 Assembly Programs

This is the basic structure of assembly codes.

```
1  MODEL SMALL
2  ;This is a comment
3  .STACK 100H
4  .DATA
5  ;data definitions go here
6  .CODE:
7  MAIN PROC
8  ;instructions go here
9  MAIN ENDP
10 ;other proced
11 END MAIN
```

2.1.1 Taking input from user

```
1  .MODEL SMALL
2  .STACK 100H
3
4  .CODE
5      MAIN PROC
6          MOV AH, 1                ; read a character
7          INT 21H
8
9          MOV BL, AL              ; save input character into BL
10
11         MOV AH, 2
12         MOV DL, 0DH
13         INT 21H
14         MOV DL, 0AH              ;line 11 to line 14 is used for printing new
            line.
15
16         MOV AH, 2                ; display the character stored in BL
17         MOV DL, BL
18         INT 21H
19
20         MOV AH, 4CH              ; return control to DOS
21         INT 21H
```

```
22  
23     MAIN ENDP  
24 END MAIN
```

3 Multiplication of two numbers(2 and 5)

```
1  TITLE PGML_1: Sample program  
2  .MODEL SMALL  
3  .STACK 100H  
4  .DATA  
5  A DW 2  
6  B DW 5  
7  RESULT DW ?  
8  .CODE  
9  MAIN PROC  
10     MOV AX,@DATA  
11     MOV DS,AX  
12     ;MUL TWO NUMBERS  
13  
14     MOV AX,A  
15     MUL B  
16     MOV RESULT,AX  
17  
18  
19     MOV AX,02H  
20     INT 21H  
21  
22  
23     MAIN ENDP  
24     END MAIN
```

4 Ouput

10

5 Discussion & Conclusion

Based on the focused objective(s) to understand about the structure of assembly language, various arithmetic operations and the additional lab exercise made me more confident towards the fulfilment of the objectives(s).

6 Lab Task (Please implement yourself and show the output to the instructor)

1. Perform addition, subtraction, multiplication and division of 6 and 2 these two numbers in a single code.
2. Take two single digit integer inputs from and perform addition, subtraction, multiplication and division on two numbers.

7 Lab Exercise (Submit as a report)

- Take a double digit number input from the user.

-
- Convert 260°C to Fahrenheit using the following expression and store in a F variable: $^{\circ}\text{F} = ^{\circ}\text{C} \times 9/5 + 32$
- 1
 - Convert 1000 °F (Fahrenheit) to °C (Celsius) using the following expression and store in a C variable: $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9 + 1$

8 Policy

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